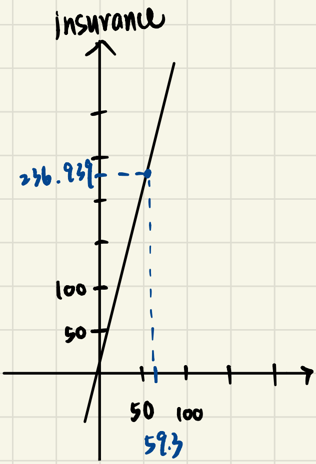
3.18

(a)

$$\widehat{insurance} = 6.855 + 3.880 \times income$$



$\overline{income} = 59.3$

$\Rightarrow \overline{insurance} = 6.855 + 59.3 \times 3.88 = 236.939$

- Slope (3.880): When income increase 1 thousand of dollars, the expected insurance increase 3.880 thousand of dollars.
- Intercept (6.855): When income is 0, the expected insurance is 6.855 thousand of dollars.

(b)

The change in insurance for a \$1,000 increase in income is $b_2 = 3.880$

95% confidence interval for β_2 : $b_2 \pm t_{0.025, 18} \cdot se(b_2)$

$\Rightarrow 3.880 \pm 2.101 \times 0.112$

$\Rightarrow [\ 3.645,\ 4.115\ ]$

We are 95% confidient that the true increase is between \$3,645 and \$4,115.

c)

$$\hat{y}_0 = 6.855 + 3.880 \times 100 = 394.855$$

$$SE(\hat{y}_0) = \sqrt{Var(b_1) + X_0^2 Var(b_2) + 2X_0 Cov(b_1, b_2)}$$

$$= \sqrt{7.383^2 + 100^2 \times 0.112^2 + 2 \times 100 \times (-0.746)}$$

$$= 5.505$$

99% C.I. for $X = 10$: $\hat{y}_0 \pm t_{0.005,18} \times se(\hat{y}_0)$

$\Rightarrow 394.855 \pm 2.878 \times 5.505$

$\Rightarrow [379.012, 410.698]$

d)

$$\begin{cases} H_0: \beta_2 = 5 \\ H_1: \beta_2 \neq 5 \end{cases}$$

Test statistic: $T = \dfrac{b_2 - 5}{se(b_2)} = \dfrac{3.880 - 5}{0.112} = -10$

Rejection region $= \{|t| \geq t_{0.025,18} = 2.101\}$

$\Rightarrow T \in$ Rejection region, we reject H_0.

At the 5% significance level, we conclude that the expected increase is significantly different from \$5,000.

(e)

$$\begin{cases} H_0: \beta_2 \le 1 \\ H_1: \beta_2 > 1 \end{cases}$$

Test statistic : $T = \frac{3.88 - 1}{0.112} = 25.71$

$R.R. = \{ t > t_{0.05,18} = 1.734 \}$

$\Rightarrow T \in R.R.$, we reject H_0

The evidence suggests that for every dollars increase in income, the increase in life insurance holdings is significantly greater than one dollar.

3.23

(a) Margin effect $= \dfrac{d(Price)}{d(sqft)} = 2 \times \alpha_2 \times sqft$

```
--- SQFT = 20 下的檢定結果 ---
邊際效應估計值: 7.38076
t 統計量: -26.72875
臨界值 (t_c): 1.647919
p-value: 1$t
```

$\begin{cases} H_0: ME \le 13 \\ H_a: ME > 13 \end{cases}$, test statistic : $t = -26.72875$

$RR = \{ T > 1.647919 \}$

$t \notin RR$ & p-value $= 1$, do not reject H_0

We don't have enough evidence to say that margin effect in SQFT = 20 is larger than thirteen.

(b)

```
--- SQFT = 40 下的檢定結果 ---
邊際效應估計值: 14.76152
t 統計量: 4.189467
臨界值 (t_c): 1.647919
p-value: 1.65395e-05$t
```

$\begin{cases} H_0: ME \le 13 \\ H_a: ME > 13 \end{cases}$, test statistic : $t = 14.76152$

$RR = \{ T > 1.647919 \}$

$t \in RR$ & p-value < 0.05, reject H_0

We have enough evidence that for 4000-square-foot house, the marginal effect of adding 100 square feet is significantly greater than $13,000

(c)

```
SQFT=20 的期望價格預測與 95% 信賴區間：
       fit      lwr      upr
1 167.3735 158.0481 176.6988
```

We are 95% confident that the averages selling price of all 2000-square-foot homes in the area will fall within this range

(d)

```
樣本中 SQFT=20 的平均售價: 163.3333
```

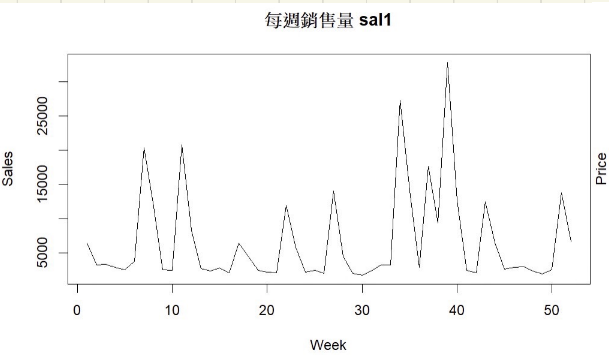
The calculated averages selling price of all homes in the sample with SQFT = 20 is $163.\overline{3}$. Comparing this value with the expected value of 167.37 predicted by the model in (c), we find that the sample mean falls within the 95% C.I : [158.50, 176.70]

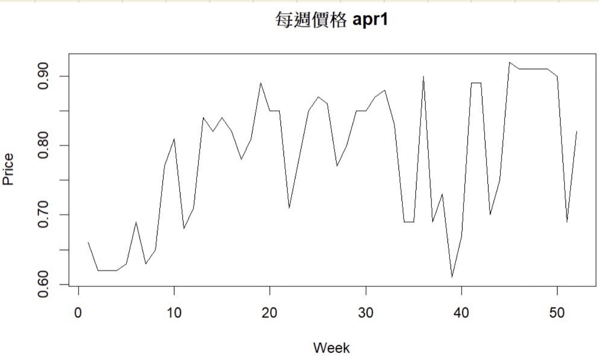
This shows that the regression model's predicted at SQFT = 20 is consistent with the actual observed data. If the sample mean falls within the interval, it mean that the model effectively represents the market average for the area.

3.41

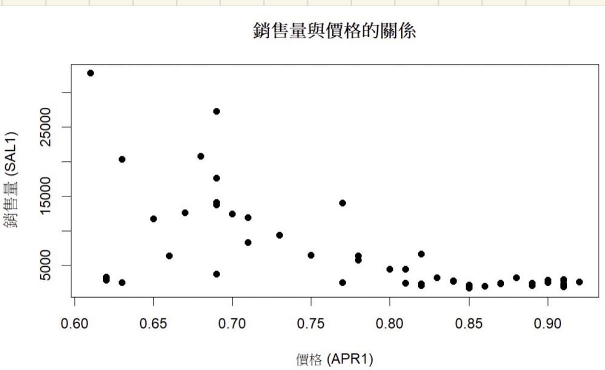
(a)

```
      sal1              apr1
 Min.   : 1762   Min.   :0.6100
 1st Qu.: 2485   1st Qu.:0.6900
 Median : 3136   Median :0.8100
 Mean   : 6719   Mean   :0.7825
 3rd Qu.: 8612   3rd Qu.:0.8625
 Max.   :32820   Max.   :0.9200
```





(b)



The inverse relationship is what I expected.

In econometrics, the demand curve is negative slope.

(c)

```
Call:
lm(formula = sal1 ~ price1, data = tuna)

Residuals:
     Min       1Q   Median       3Q      Max 
-10869.5  -1588.3   -499.3   1626.2  18607.1 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45      78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,	Adjusted R-squared:  0.367 
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06

           2.5 %    97.5 %
price1 -592.2897 -276.6049
```

b_2 represents that the number of tuna cans sold for decreasing 434.45 units when every 1-cent increase in price.

d)

```
       fit      lwr      upr
1 10302.9 8624.934 11980.87
```

$\Rightarrow$ 90% C.I. : [8624.934, 11980.87]

(e)

```
平均值處的彈性: -5.059825
彈性的 95% 信賴區間: -6.898149 -3.221501
```

The forumla for price elasticity ε in a linear model is: $b_2 \times \frac{\overline{PRICE1}}{\overline{SAL1}}$

This value tells us the percentage change in quantity demanded for a 1% change in price.

Since $\overline{PRICE1}$ and $\overline{SAL1}$ are treated as constants, the interval is

[Lower Bound of $b_2 \times \frac{\overline{P}}{\overline{S}}$, Upper Bound of $b_2 \times \frac{\overline{P}}{\overline{S}}$]

Since $|\varepsilon| \approx 5 > 1$, the demand for tuna is price elastic. This indicates that consumers are highly response to price changes.

(f)

檢定 H0: epsilon = -3 的 t 統計量: -2.250572 p 值: 0.02884204

$$\begin{cases} H_0: \varepsilon = -3 \\ H_1: \varepsilon \neq -3 \end{cases} \longrightarrow \begin{cases} H_0: \beta_2 = -3 \times \dfrac{\bar{S}}{\bar{P}} \\ H_1: \beta_2 \neq -3 \times \dfrac{\bar{S}}{\bar{P}} \end{cases}$$

test statistic : $t = \dfrac{b_2 - \beta_{2,H_0}}{SE(b_2)} = -2.250572$

$RR = \{ |T| \geq t_{0.025,50} = 2.009 \}$, $t \in RR$.

$\Rightarrow$ reject H_0 , the tuna market's price sensitivity is statistically different from -5